I. Heat transfer

Symbology

Symbol	Quantity	Unit
λ , k	Thermal conductivity	$\mathrm{W/(m \cdot K)}$
$\dot{q}$	Heat flux	$\mathrm{W/m^2}$
$\dot{Q}$	Heat flow	W
α	Heat transfer coefficient	$\mathrm{W/(m^2 \cdot K)}$
a	Thermal diffusivity	$\mathrm{m^2/s}$
ν	Kinematic viscosity	$\mathrm{m^2/s}$

1. Types of heat transfer

1.1 Heat conduction

Occurs with all materials when there is a temperature gradient.

Steady-state conduction (1st Fourier's law):

$$\dot{q} = -\lambda \frac{\mathrm{d}T}{\mathrm{d}x}$$

Transient conduction (2nd Fourier's law 1D):

$$\frac{\partial T}{\partial t} = \frac{\lambda}{\rho c_p} \frac{\partial^2 T}{\partial x^2}$$

1.2 Convective heat transfer

This occurs in moved fluids.

Newton's approach:

$$\dot{q} = \alpha \left(T_1 - T_2\right) = \frac{\lambda_{\text{fluid}}}{\delta_{\text{th}}} \left(T_1 - T_2\right)$$

1.3 Radiation

Bodies with a temperature above 0 K emit thermal radiation.

2. Heat conduction and conductivity

2.1 Heat flux

$$\dot{q} = -\lambda \frac{\mathrm{d}T}{\mathrm{d}x} = k\Delta T = \frac{\lambda}{\delta}\Delta T$$

Procedure:

$$\vec{\dot{q}} \int_0^{\delta} dx = -\lambda \int_{T_1}^{T_2} dT \Longrightarrow \dot{q} = \frac{\lambda}{\delta}\left(T_1 - T_2\right)$$

2.2 Heat flow

$$\dot{Q} = \frac{\Delta Q}{\Delta t} = \dot{q}A = kA\Delta T = \frac{\lambda}{\delta} A \Delta T$$

2.3 Thermal conduction k^{-1} and Thermal conductivity k

$$k_{\text{tot}}^{-1} = \sum_{i=1}^{n} \frac{\delta_i}{\lambda_i} \implies k_{\text{tot}} = \frac{1}{k_{\text{tot}}^{-1}}$$

3. Heat conductivity through a hollow cylinder

$$\dot{Q} = -\lambda A \frac{\mathrm{d}T}{\mathrm{d}r} = -\lambda 2\pi r l \frac{\mathrm{d}T}{\mathrm{d}r}$$

$$\boxed{\dot{Q} = 2\pi l \cdot \frac{T_0 - T_1}{\frac{1}{\lambda} \ln \frac{r_1}{r_0}} = 2\pi r_0 l k \Delta T}$$

With multilayered walls:

$$\dot{Q} = 2\pi l \cdot \frac{T_0 - T_n}{\sum_{i=1}^{n} \frac{1}{\lambda_i} \ln \frac{r_i}{r_{i-1}}} = \boxed{2\pi r_0 l k_0 (T_0 - T_n)}$$

3.1 Thermal conductivity k

$$k_0 = \frac{1}{r_0 \left(\left(\frac{1}{\lambda_1} \ln \frac{r_1}{r_0}\right) + \left(\frac{1}{\lambda_2} \ln \frac{r_2}{r_1}\right) + \cdots + \left(\frac{1}{\lambda_n} \ln \frac{r_n}{r_{n-1}}\right)\right)}$$

$$k_0^{-1} = r_0 \sum_n \left(\frac{1}{\lambda_n} \ln \frac{r_n}{r_{n-1}}\right)$$

3.2 Thin-walled pipe approximated as a flat wall

$$\dot{Q}_T = \lambda A \frac{T_0 - T_1}{\delta} \quad ; \quad A = 2\pi l \left(r_0 + \frac{\delta}{2}\right)$$

$$\frac{\dot{Q}_T}{\dot{Q}_W} = \frac{\delta}{\left(r_0 + \frac{\delta}{2}\right) \ln \left(1 + \frac{\delta}{r_0}\right)} = \frac{\delta}{r_0} \frac{1}{\left(1 + \frac{\delta}{2r_0}\right) \ln \left(1 + \frac{\delta}{r_0}\right)}$$

4. Convective heat transfer

4.1 Reynold's number

It compares inertial to viscous forces.

$$\boxed{Re = \rho\, c \frac{L_{\text{char}}}{\eta} = c \frac{L_{\text{char}}}{\nu}}$$

$Re < 2300$: Laminar flow;
$Re \approx 2300$: Transition zone;
$Re > 2300$: Turbulent flow

4.2 Prandtl number

It compares momemtum diffusivity to thermal diffusivity.
Sets velocity's relative thickness and thermal boundary layers.

$$Pr = \frac{\nu}{a} = \frac{\delta_{\text{fluid}}}{\delta_{\text{th}}} = \frac{\eta c_p}{\lambda} \quad ; \quad a = \frac{\lambda}{\rho c_p} \quad ; \quad \nu = \frac{\eta}{\rho}$$

4.3 Nusselt number

Measures convective heat transfer relative to pure conduction across a boundary layer and the heat transfer coefficient α.

$$Nu = \frac{\alpha \cdot L_{\text{char}}}{\lambda_{\text{fl}}} = \dot{q} \frac{L_{\text{char}}}{\lambda \Delta T_{\text{char}}}$$

$$\alpha = \frac{Nu \cdot \lambda}{L_{\text{char}}}$$

4.3.1 Forced convection

$$\boxed{Nu = c \cdot Re^m \cdot Pr^n}$$

where c is a constant.

For laminar flow over a flat plate:

$$Nu = 0.664 \cdot \sqrt{Re} \cdot \sqrt[3]{Pr}$$

For a flow in the transition zone (Tz):

$$Nu_{\text{Tz}} = \sqrt{Nu_{\text{lam}}^2 + Nu_{\text{turb}}^2}$$

4.4 Grashof number

It compares buoyancy to viscous forces.

$$Gr = g\beta\Delta T \frac{L^3}{\nu^2} \quad ; \quad g = 9.81 \text{ N/kg}$$

$$\beta\Delta T = \frac{\rho_\infty - \rho_0}{\rho_\infty} = \frac{\Delta\rho}{\rho_\infty}$$

$$Gr = g \frac{\rho_\infty - \rho_0}{\rho_\infty} \frac{L^3}{\nu^2}$$

4.4.1 Natural convection

Peclet number

It compares advective to diffusive heat transport, i.e., the strength of convection relative to thermal diffusion.

$$Pe = RePr$$

Rayleigh number

It combines buoyancy driving and thermal diffusion, and it measures the intensity of natural convection.

$$Ra = GrPr$$

5. Heat exchanger

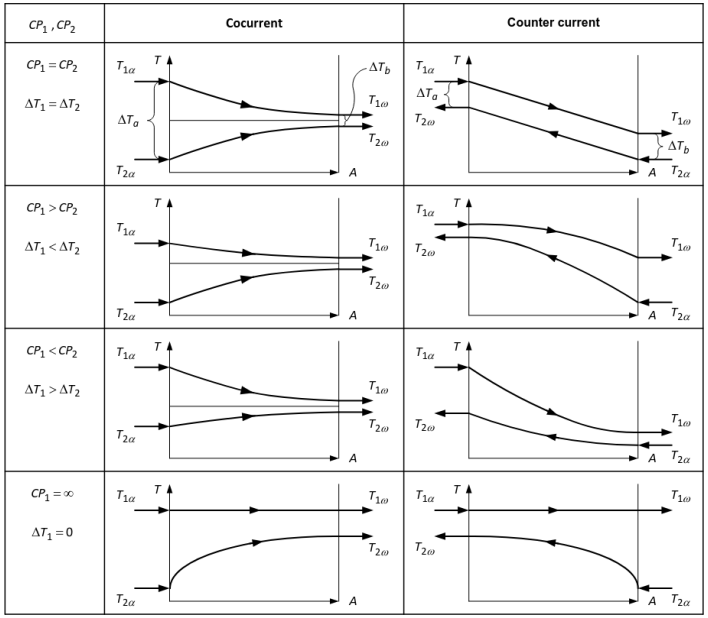
5.1 Energy balance

$$\dot{Q} = \dot{m} c_p \Delta\vartheta = CP\Delta\vartheta$$

$$\dot{Q}_1 = \dot{Q}_2$$

$$\dot{m}_1 c_{p1} \Delta\vartheta_1 = \dot{m}_2 c_{p2} \Delta\vartheta_2 \Longleftrightarrow CP_1 \Delta\vartheta_1 = CP_2 \Delta\vartheta_2$$

5.2 Average temperature difference



5.2.1 Logarithmic mean temperature difference (LMTD)

$$\Delta T_m = \frac{\Delta T_a - \Delta T_b}{\ln \dfrac{\Delta T_a}{\Delta T_b}}$$

where:

$$\Delta T_a = |\vartheta_{1,a} - \vartheta_{2,a}| \quad ; \quad \Delta T_b = |\vartheta_{1,b} - \vartheta_{2,b}|$$

5.2.2 Mean temperature heat flow

$$\dot{Q} = kA\Delta T_m$$

5.2.3 Heat capacity rate ratio (Capacity ration)

It measures the relative ability of stream 1 vs. stream 2 to carry heat per kelvin:

$$R_1 = \frac{\Delta\vartheta_1}{\Delta\vartheta_2} = \frac{CP_1}{CP_2}$$

5.3 Cocurrent vs countercurrent flow heat exchanger

$$\dot{Q} = kA_{\text{co}}\Delta T_{m,\text{co}} = kA_{\text{count}}\Delta T_{m,\text{count}}$$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\Delta T_{m,\text{count}}}{\Delta T_{m,\text{co}}}$$

5.3.1 If $\dot{Q}_1 \neq \dot{Q}_2$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\dot{Q}_{\text{count}}\Delta T_{m,\text{count}}}{\dot{Q}_{\text{co}}\Delta T_{m,\text{co}}}$$

II. The 2nd law of thermodynamics

6. Recap of the 1st law

6.1 Closed systems

$$q_{12} + w_{12} = u_2 - u_1$$

6.2 Open systems

$$q_{12} + w_{12} = h_2 - h_1 + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$w_{t12} = \int v\,dp + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right) + j_{12}$$

$$q_{12} + \int v\,dp + j_{12} = h_2 - h_1 \overset{\text{ideal gas}}{=\!=\!=} c_p\left(T_2 - T_1\right)$$

7. The 2nd law

7.1 Statements

$$\boxed{\text{Energy} = \text{Exergy} + \text{Anergy}}$$

Processes	Reversible	Irreversible
Energy	Conserved	Reduced → Loss
Entropy	Constant	Increases → Production

7.2 Entropy

Entropy S: extensive state variable [J/K]

Specific entropy s: intensive state variable [J/kgK]

Entropy flow $\dot{S}$: entropy over time [W/K]

8. Entropy balance equation

$$\boxed{dS = dS_Q + dS_m + dS_{irr}}$$

8.1 Entropy balance terms

- Heat transfer across the system boundary:

$$dS_Q = \frac{dQ}{T}$$

- Mass flow across the system boundary:

$$dS_{\text{m}} = \dot{S} = \dot{m}\,s$$

- Irreversible processes inside the system:

$$\dot{S}_{irr}(t)\begin{cases} dS_{irr} > 0 & \text{irreversible processes} \\ dS_{irr} = 0 & \text{reversible processes} \\ dS_{irr} < 0 & \text{impossible} \end{cases}$$

8.2 Entropy balance equation for closed systems

$$dS = dS_Q + \cancelto{0}{dS_m} + dS_{irr}$$

$$dS = \dot{S}_Q\,dt + \dot{S}_{irr}\,dt = \left(\frac{\dot{Q}}{T} + \dot{S}_{irr}\right)dt$$

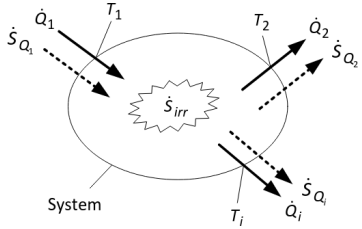
8.2.1 Closed system entropy flow balance equation

$$\frac{dS}{dt} = \sum_i \dot{S}_{Qi} + \dot{S}_{irr} \quad ; \quad \dot{S}_{irr} \geq 0$$

For steady-state case:

$$\dot{S}_{irr} = -\sum_i \dot{S}_{Qi} = -\sum_i \frac{\dot{Q}_i}{T_i} \geq 0$$

8.2.2 Visual representation



$$\boxed{\sum_i \dot{S}_{Qi} = \sum_i \frac{\dot{Q}_i}{T_i} = -\dot{S}_{irr}}$$

8.3 Entropy balance equation for open systems

$$\frac{dS}{dt} = \sum_\alpha \dot{m}_\alpha s_\alpha - \sum_\omega \dot{m}_\omega s_\omega + \sum \dot{S}_Q(t) + \dot{S}_{irr}(t)$$

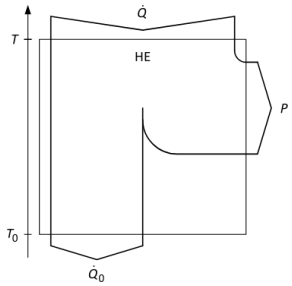
8.3.1 Steady-state flow processes

$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha - \sum \dot{S}_Q \geq 0$$

8.3.2 Steady-state and adiabatic flow processes

$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha \geq 0$$

9. The thermal enigne



$$\dot{Q} + \dot{Q}_0 + P = 0$$

$$\eta_{th} = \frac{-P}{\dot{Q}} = 1 - \frac{\left|\dot{Q}_0\right|}{\dot{Q}}$$

$$\dot{S}_{irr} = -\sum_{i=1}^{n} \dot{S}_{Qi} = -\frac{\dot{Q}}{T} - \frac{\dot{Q}_0}{T_0} \geq 0$$

$$-\dot{Q}_0 = T_0\left(\frac{\dot{Q}}{T} + \dot{S}_{irr}\right)$$

9.1 Carnot efficiency

$$\eta_{th} = 1 - \frac{T_0}{T} - \cancelto{0,\ \text{ideal}}{\frac{T_0\dot{S}_{irr}}{\dot{Q}}}$$

$$\eta_{th,max} = \eta_C(T, T_0) = 1 - \frac{T_0}{T}$$

10. Entropy S as a state variable

$$ds = \frac{dq + dj}{T} = \frac{du + p\,dv}{T} = \frac{dh - v\,dp}{T}$$

$$T\,ds = dq + dj \Longrightarrow T\,(s_2 - s_1) = q_{12} + j_{12}$$

10.1 Reversible processes

10.1.1 Clausius entropy relation

$$\boxed{\delta q_{\text{rev}} = T\,ds \Longleftrightarrow \delta Q_{\text{rev}} = T\,dS}$$

10.1.2 Work W and heat Q

$$w_{p,\,rev} = \int v\,dp$$

$$Q_{rev} = \int T\,dS$$

10.2 Dissipation j as a state variable

10.2.1 Dissipation

$$w_{t12} = w_{p12} + j_{12} \Longleftrightarrow w_{p12} = w_{t12} - j_{12}$$

$$dq_{rev} + dw_{v,\,rev} = du$$

$$T\,ds = dq + T\,ds_{irr} = dq_{rev} + dj$$

$$\int_1^2 T\,ds = q_{12} + j_{12}$$

10.2.2 For closed systems

$$ds = \frac{du + p\,dv}{T} = \frac{d_v\,dT}{T} + \frac{R_i\,dv}{v}$$

$$s_2 - s_1 = c_v \ln\frac{T_2}{T_1} + R_i \ln\frac{v_2}{v_1}$$

10.2.3 Work change

All the works can be calculated with a temperature change:

$$q_{12} + w_{t12} = h_2 - h_1 + \frac{c_2^2 - c_1^2}{2} + g\,(z_2 - z_1)$$

$$q_{12} + \int v\,dp + j_{12} = h_2 - h_1 = c_p\,(T_2 - T_1)$$

$$\int T\,ds = q_{12} + j_{12} = \boxed{\int T\,ds + \int v\,dp = h_2 - h_1}$$

$$\boxed{T\,ds + v\,dp = h_2 - h_1 \Longleftrightarrow T\,ds - p\,dv = u_2 - u_1}$$

10.3 Reversible state changes with perfect gases

10.3.1 Open system

$$q_{12} + j_{12} + w_{p,rev} = h_2 - h_1$$

$$\int_1^2 T\,ds + \int_1^2 v\,dp = c_p\,(T_2 - T_1)$$

10.3.2 Closed system

$$q_{12} + j_{12} + w_{p,rev} = u_2 - u_1$$

$$\int_1^2 T\,ds - \int_1^2 p\,dv = c_v\,(T_2 - T_1)$$

10.4 Polytropic state changes

$$T\,ds = dq + dj = du + p\,dv = dh - v\,dp$$

$$du + p\,dv = c_v\,dT + p\,dv \quad ; \quad dh - v\,dp = c_p\,dT - v\,dp$$

$$dw_v = c_v\frac{k-1}{n-1}\,dT = -p\,dv$$

For the change in entropy of the polytropic state change:

$$ds = c_v\frac{n-k}{n-1}\frac{dT}{T}$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{n-1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$

$$\boxed{s_2 - s_1 = c_v\frac{n-k}{n-1}\ln\frac{T_2}{T_1}} \quad n \neq 1$$

$$\boxed{s_2 - s_1 = c_v \ln\frac{T_2}{T_1} + R_i \ln\frac{v_2}{v_1} = c_p \ln\frac{T_2}{T_1} - R_i \ln\frac{p_2}{p_1}}$$

Compressor with dissipation: $n > k$
Turbine with dissipation: $n < k$

11. Exergy and anergy

Exergy: useful part of energy that can be completely converted into work under given ambient conditions.

Anergy: remaining part of energy that cannot be converted into work under given ambient conditions.

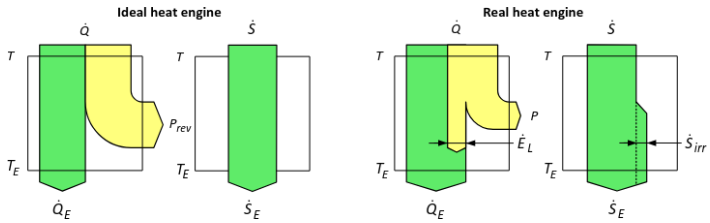
$$\boxed{e = ex + an}$$

11.1 Statements of the 2nd LT

I. In all irreversible processes exergy is converted in anergy
II. Only in reversible processes exergy remains constant
III. It is impossible to convert anergy into exergy

11.2 Exergy

11.2.1 Exergy balance / Exergy loss



$$\dot{E}_L = \sum_\alpha \dot{E}_\alpha - \sum_\omega |\dot{E}_\omega| = \dot{Q}\frac{T_A - T_E}{T_A} + \left(-\dot{Q}\frac{T_B - T_E}{T_B}\right)$$

$$\boxed{\dot{E}_L = T_E\dot{S}_{irr} = T_E\frac{T_A - T_B}{T_A T_B}\dot{Q}}$$

$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha - \sum \dot{S}_Q \geq 0, \quad d\dot{S}_Q = \frac{d\dot{Q}}{T}$$

11.2.2 Exergy of heat

$$\dot{E}_Q = P_{rev} = \left(1 - \frac{T_E}{T}\right)\dot{Q} = \eta_C\dot{Q}$$

11.2.3 Exergy of material flows (without mixtures)

$$e = h - h_E - T_E\,(s - s_E) + \frac{c^2}{2} + gz + q\left(1 - \frac{T_E}{T}\right)$$

$$e_2 - e_1 = h_2 - h_1 - T_E\,(s_2 - s_1)$$

Perfect gases

$$e = c_p\left(T - T_E - T_E\ln\frac{T}{T_E}\right) + R_i T_E \ln\frac{p}{p_E}$$

Incompressible liquids (isobaric)

$$e = c_p\left(T - T_E - T_E\ln\frac{T}{T_E}\right)$$

Isobaric for $T < T_E$ (cooling)

$$e = c_p\left(T_E - T - T_E\ln\frac{T_E}{T}\right)$$

III. Steady-state flow processes

12. Adiabatic working cycles

12.1 Compressor

12.1.1 1st law

$$w_{t12} + q_{12} = h_2 - h_1 + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$w_{t12} + q_{12} = c_p\left(T_2 - T_1\right) + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$P_{12} + \dot{Q}_{12} = \dot{m}\left(w_{t12} + q_{12}\right)$$

12.1.2 Isentropic compression efficiency factor

$$\eta_s = \frac{\text{isentropic work}}{\text{real polytropic work}}$$

$$\eta_s = \frac{\Delta h_s}{\Delta h} = \frac{h_{2,s} - h_1}{h_2 - h_1} = \frac{c_{p,m}\left(T_{2,s} - T_1\right)}{c_{p,m}\left(T_2 - T_1\right)} = \frac{T_{2,s} - T_1}{T_2 - T_1} \leq 1$$

12.2 Turbine

12.2.1 1st law

$$w_{t12} + q_{12} = h_2 - h_1 + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$w_{t12} + q_{12} = c_p\left(T_2 - T_1\right) + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$P_{12} = \dot{m} w_{t12} \iff -P_{12} = \dot{m}\left|w_{t12}\right|$$

12.2.2 Isentropic turbine efficiency

$$\eta_s = \frac{\text{real polytropic work}}{\text{isentropic work}}$$

$$\eta_s = \frac{-\Delta h}{-\Delta h_s} = \frac{h_1 - h_2}{h_1 - h_{2,s}} = \frac{T_1 - T_2}{T_1 - T_{2,s}}$$

13. Adiabatic flow processes without work

13.1 Throttle / Expansion valve (isenthalpic)

$$h_1 = h_2 \Longrightarrow j_{12} = -\int_1^2 v\, dp = -R_i T \ln \frac{p_2}{p_1}$$

13.2 Nozzle

$$h_2^0 = h_1^0 \Longrightarrow h_2 + \frac{1}{2}c_2^2 = h_1 + \frac{1}{2}c_1^2$$

$$c_2 = \sqrt{2\left(h_1 - h_2\right) + c_1^2}$$

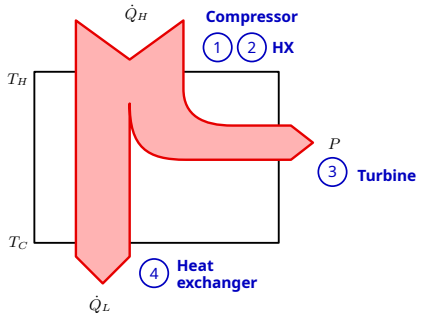
where h^0 is the total enthalpy: $h^0 = h + \frac{c^2}{2}$

IV. Thermodynamics cycles

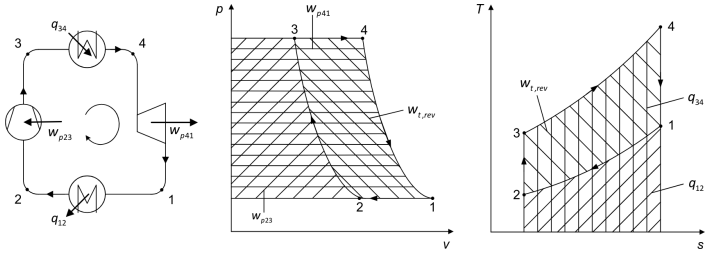
14. Reversible cycle work

Total cycle / sum of all sub-processes:

$$\boxed{w_{t,\,rev} = \int_1^2 v\, dp + \int_2^3 v\, dp + \ldots + \int_n^1 v\, dp = \oint v\, dv = \sum_{i=1}^{n} w_{pi}}$$



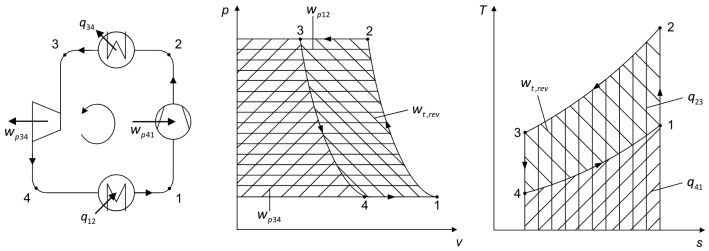
14.1 Clockwise cycle



$$w_{t,rev} = w_{p,23} + w_{p,41} < 0$$

$$\left|w_{t,rev}\right| = \left|q_{34}\right| - \left|q_{12}\right|$$

14.2 Counterclockwise cycle

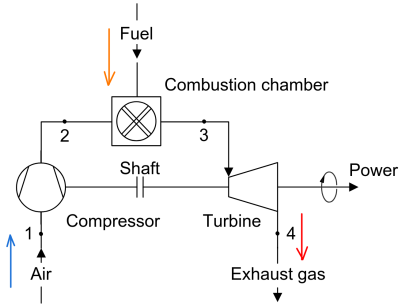


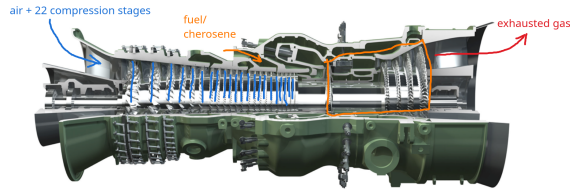
$$w_{t,rev} = w_{p,12} + w_{p,34} < 0$$

$$\left|w_{t,rev}\right| = \left|q_{23}\right| - \left|q_{41}\right|$$

V. Thermal systems with gas turbines

15. Turbine





Power balance:

$$\dot{Q}_{23} + \dot{Q}_{41} + P_C = 0$$

15.1 Joule process

$$\pi = \frac{p_2}{p_1} = \frac{p_3}{p_4}$$

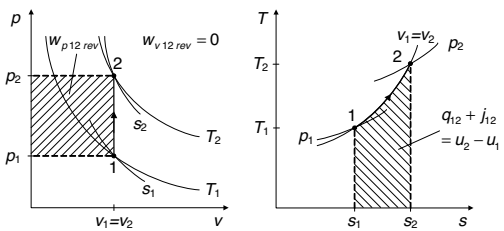
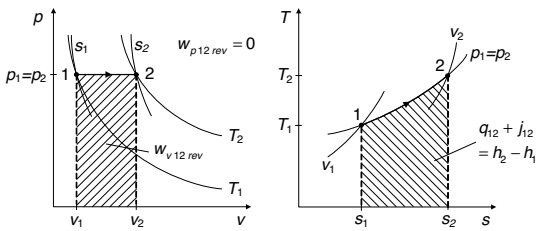
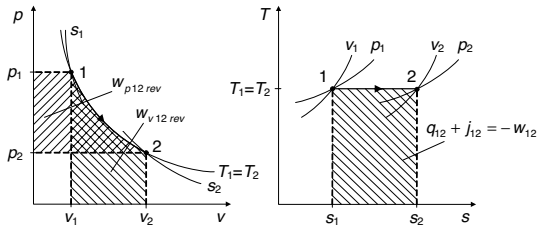
$$-w_C = q_\alpha + q_\omega = q_{23} + q_{41} = c_p\left(T_3 - T_2 + T_1 - T_4\right)$$

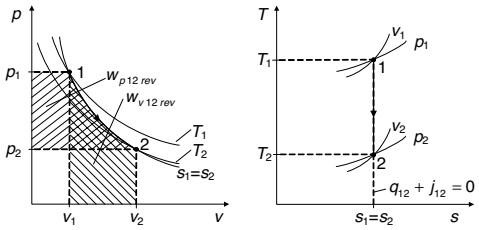
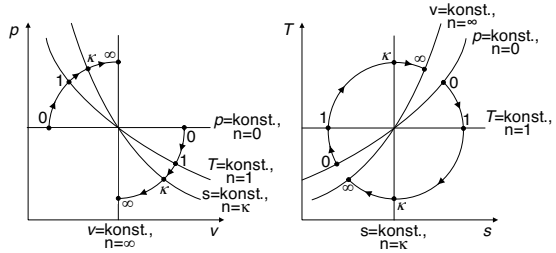
$$\left|w_C\right| = c_p\left(T_3 - T_2\right)\left(1 - \frac{T_4 - T_1}{T_3 - T_2}\right)$$

$$\left(1 - \frac{T_4 - T_1}{T_3 - T_2}\right) = \left(\frac{p}{p_0}\right)^{\frac{k-1}{k}} = \left(\frac{1}{\pi}\right)^{\frac{k-1}{k}}$$

$$\eta_J = \frac{\left|w_C\right|}{q_\alpha} = 1 - \frac{T_1}{T_2} = 1 - \left(\frac{1}{\pi}\right)^{\frac{k-1}{k}}$$

Special state changes with perfect gases (without consideration of kinetic and potential energies)

Ideal gas	$v =$ **constant: Isochoric state change**	$p =$ **constant: Isobaric state change**	$T =$ **constant: Isothermal state change**
Therm. state equation $p\,V = n\,R\,T$ $R = 8.314\,\mathrm{J/mol\,K}$ $p\,V = m\,R_i\,T$ $R_i = \dfrac{R}{M_i}$ $p\,v = R_i\,T, \quad \rho = \dfrac{p}{R_i\,T}$ $R_i = c_p - c_v, \quad \kappa = \dfrac{c_p}{c_v}$ $c_v = \dfrac{R_i}{\kappa - 1}, \quad c_p = \dfrac{\kappa\,R_i}{\kappa - 1}$	 $v = \text{const.}: \quad \dfrac{p_1}{T_1} = \dfrac{p_2}{T_2} = \dfrac{p}{T} = \text{const.}$ $w_{v\,12\,rev} = 0 \quad (\mathrm{d}v = 0)$ $w_{p\,12\,rev} = v\,(p_2 - p_1) = \dfrac{p_2 - p_1}{\rho}$	 $p = \text{const.}: \quad \dfrac{v_1}{T_1} = \dfrac{v_2}{T_2} = \dfrac{v}{T} = \text{const.}$ $w_{v\,12\,rev} = -p\,(v_2 - v_1) = -R_i\,(T_2 - T_1)$ $w_{p\,12\,rev} = 0 \quad (\mathrm{d}p = 0)$	 $T = \text{const.}: \quad p_1\,v_1 = p_2\,v_2 = p\,v = \text{const.}$ $w_{v\,12\,rev} = -p_1\,v_1\,\ln\dfrac{v_2}{v_1} = \underbrace{p_1\,v_1\,\ln\dfrac{p_2}{p_1}}_{[\mathrm{J}]} = \underbrace{R_i\,T\,\ln\dfrac{p_2}{p_1}}_{[\mathrm{J/kg}]}$ $w_{p\,12\,rev} = w_{v\,12\,rev}$
1st LT closed system $q_{12} + w_{12} = u_2 - u_1$ Rev. volume change work: $w_{v\,12\,rev} = -\int_1^2 p\,\mathrm{d}v$ $w_{v\,12} = w_{v\,12\,rev} + j_{12}$ Overall work: $w_{12} = w_{v\,12} + w_{W\,12} + w_{el\,12}$	$q_{12} + w_{12} = u_2 - u_1$ Rev. SC: $w_{12} = w_{v\,12\,rev} = 0 \quad (\mathrm{d}v = 0)$ $q_{12\,rev} = u_2 - u_1 = c_v\,(T_2 - T_1)$	$q_{12} + w_{12} = u_2 - u_1$ Rev. SC: $w_{12} = w_{v\,12\,rev} = -p\,(v_2 - v_1)$ $q_{12\,rev} = u_2 - u_1 + p\,(v_2 - v_1) = h_2 - h_1$ $q_{12\,rev} = c_p\,(T_2 - T_1)$	$q_{12} + w_{12} = u_2 - u_1 = c_v\,(T_2 - T_1) = 0$ $q_{12} = -w_{12}$ Rev. SC: $q_{12\,rev} = -w_{v\,12\,rev}$ $q_{12\,rev} = p_1\,v_1\,\ln\dfrac{v_2}{v_1} = -p_1\,v_1\,\ln\dfrac{p_2}{p_1} = -R_i\,T\,\ln\dfrac{p_2}{p_1}$
1st LT open system $q_{12} + w_{t\,12} = h_2 - h_1$ Rev. pressure change work: $w_{p\,12\,rev} = \int_1^2 v\,\mathrm{d}p$ Technical work: $w_{t\,12} = \int_1^2 v\,\mathrm{d}p + j_{12}$	$q_{12} + w_{t\,12} = h_2 - h_1$ Rev. SC: $w_{t\,12} = w_{p\,12\,rev} = v\,(p_2 - p_1)$ $q_{12\,rev} = h_2 - h_1 - v\,(p_2 - p_1) = u_2 - u_1$ $q_{12\,rev} = c_v\,(T_2 - T_1)$	$q_{12} + w_{t\,12} = h_2 - h_1$ Rev. SC: $w_{t\,12} = w_{p\,12\,rev} = 0 \quad (\mathrm{d}p = 0)$ $q_{12\,rev} = h_2 - h_1 = c_p\,(T_2 - T_1)$	$q_{12} + w_{t\,12} = h_2 - h_1 = c_p\,(T_2 - T_1) = 0$ $q_{12} = -w_{t\,12}$ Rev. SC: $q_{12\,rev} = -w_{p\,12\,rev}$ $q_{12\,rev} = p_1\,v_1\,\ln\dfrac{v_2}{v_1} = -p_1\,v_1\,\ln\dfrac{p_2}{p_1} = -R_i\,T\,\ln\dfrac{p_2}{p_1}$
2nd LT / entropy $\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T}$ $\mathrm{d}s = \dfrac{\mathrm{d}u + p\,\mathrm{d}v}{T} = \dfrac{\mathrm{d}h - v\,\mathrm{d}p}{T}$	$\mathrm{d}s = \dfrac{\mathrm{d}u}{T} = \dfrac{c_v\,\mathrm{d}T}{T} \Rightarrow s_2 - s_1 = c_v\,\ln\dfrac{T_2}{T_1}$	$\mathrm{d}s = \dfrac{\mathrm{d}h}{T} = \dfrac{c_p\,\mathrm{d}T}{T} \Rightarrow s_2 - s_1 = c_p\,\ln\dfrac{T_2}{T_1}$	$\mathrm{d}s = \dfrac{p\,\mathrm{d}v}{T} = \dfrac{R_i\,\mathrm{d}v}{v} \Rightarrow s_2 - s_1 = R_i\,\ln\dfrac{v_2}{v_1}$ $\mathrm{d}s = -\dfrac{v\,\mathrm{d}p}{T} = -\dfrac{R_i\,\mathrm{d}p}{p} \Rightarrow s_2 - s_1 = -R_i\,\ln\dfrac{p_2}{p_1}$

Ideal gas	$s =$ **constant: Isentropic state change**	$n =$ **constant: Polytropic state change**
Therm. state change $p\,V = n\,R\,T$ $R = 8.314\,\mathrm{J/mol\,K}$ $p\,V = m\,R_i\,T$ $R_i = \dfrac{R}{M_i}$ $p\,v = R_i\,T, \quad \rho = \dfrac{p}{R_i\,T}$ $R_i = c_p - c_v, \quad \kappa = \dfrac{c_p}{c_v}$ $c_v = \dfrac{R_i}{\kappa - 1}, \quad c_p = \dfrac{\kappa\,R_i}{\kappa - 1}$	 $s = \text{const}: \quad p_1\,v_1^\kappa = p_2\,v_2^\kappa = \text{const.} \quad \Rightarrow \quad \dfrac{T_2}{T_1} = \left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} = \left(\dfrac{v_1}{v_2}\right)^{\kappa-1}$ $w_{v\,12\,rev} = \dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \text{etc. (replace)}$ $w_{p\,12\,rev} = \kappa\,w_{v\,12\,rev} = \kappa\,\dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \text{etc. (replace)}$	 Polytrope: $pv^n = \text{konst.}$ Isobare: n=0, $pv^0 = \text{konst.}$ Isotherme: n=1, pv = konst. Isentrope: n=κ, $pv^\kappa = \text{konst.}$ Isochore: n=∞, $pv^\infty = \text{konst.}$ $n = \dfrac{\ln\frac{p_2}{p_1}}{\ln\frac{v_1}{v_2}} = \dfrac{\ln\frac{p_2}{p_1}}{\ln\frac{p_2}{p_1} - \ln\frac{T_2}{T_1}}$ $n = \text{const}: \quad p_1\,v_1^n = p_2\,v_2^n = \text{const.} \quad \Rightarrow \quad \dfrac{T_2}{T_1} = \left(\dfrac{p_2}{p_1}\right)^{\frac{n-1}{n}} = \left(\dfrac{v_1}{v_2}\right)^{n-1}$ $w_{v\,12} = \dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc. (replace, } n \neq 1)$ $w_{p\,12} = n\,w_{v\,12} = n\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc. (replace, } n \neq 1)$
1st LT closed system $q_{12} + w_{12} = u_2 - u_1$ Rev. volume change work: $w_{v\,12\,rev} = -\int_1^2 p\,\mathrm{d}v$ $w_{v\,12} = w_{v\,12\,rev} + j_{12}$ Overall work: $w_{12} = w_{v\,12} + w_{W\,12} + w_{el\,12}$	$q_{12} + w_{12} = u_2 - u_1$ Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$): $w_{12} = w_{v12\,rev} = u_2 - u_1 = c_v\,(T_2 - T_1)$ $w_{v12\,rev} = \dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \dfrac{R_i\,T_1}{\kappa - 1}\left[\left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1\right] = \text{etc.}$	$q_{12} + w_{12} = u_2 - u_1 = c_v\,(T_2 - T_1)$ $w_{12} = w_{v\,12} = \dfrac{R_i}{n - 1}\,(T_2 - T_1) = c_v\,\dfrac{\kappa - 1}{n - 1}\,(T_2 - T_1) =$ $= \dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$ Rev. SC: $w_{12} = w_{v12\,rev}$ $q_{12\,rev} = c_v\,\dfrac{n - \kappa}{n - 1}\,(T_2 - T_1) = \dfrac{n - \kappa}{\kappa - 1}\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$
1st LT open system $q_{12} + w_{t\,12} = h_2 - h_1$ Rev. pressure change work: $w_{p\,12\,rev} = \int_1^2 v\,\mathrm{d}p$ Technical work: $w_{t\,12} = \int_1^2 v\,\mathrm{d}p + j_{12}$	$q_{12} + w_{t\,12} = h_2 - h_1$ Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$): $w_{t\,12} = w_{p\,12\,rev} = \kappa\,w_{v\,12\,rev} = h_2 - h_1 = c_p\,(T_2 - T_1)$ $w_{p\,12\,rev} = \kappa\,\dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \kappa\,\dfrac{R_i\,T_1}{\kappa - 1}\left[\left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1\right] = \text{etc.}$	$q_{12} + w_{t\,12} = h_2 - h_1 = c_p\,(T_2 - T_1)$ $w_{t\,12} = w_{p\,12} = n\,w_{v\,12} = n\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$ Rev. SC: $w_{p\,12\,rev} = n\,w_{v\,12\,rev} = \dfrac{n\,R_i}{n - 1}\,(T_2 - T_1) = \text{etc.}$ $q_{12\,rev} = c_v\,\dfrac{n - \kappa}{n - 1}\,(T_2 - T_1) = \dfrac{n - \kappa}{\kappa - 1}\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$
2nd LT / entropy $\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T}$ $\mathrm{d}s = \dfrac{\mathrm{d}u + p\,\mathrm{d}v}{T} = \dfrac{\mathrm{d}h - v\,\mathrm{d}p}{T}$	$\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T} = 0 \Rightarrow s_2 - s_1 = 0$	$s_2 - s_1 = c_v\,\dfrac{n - \kappa}{n - 1}\,\ln\dfrac{T_2}{T_1} \quad (\text{for } n \neq 1)$ $s_2 - s_1 = c_v\,\ln\dfrac{T_2}{T_1} + R_i\,\ln\dfrac{v_2}{v_1} = c_p\,\ln\dfrac{T_2}{T_1} - R_i\,\ln\dfrac{p_2}{p_1}$